

EXPERIMENT 10

Aim

To implement an Autoencoder using TensorFlow/Keras to remove noise from images in the MNIST dataset.

Problem Statement

Images often contain noise which reduces quality and affects analysis.

The problem is to reconstruct clean images from noisy input images using an autoencoder model.

Dataset Link

<https://www.kaggle.com/datasets/oddrational/mnist-in-csv>

Methodology

Data Collection

The MNIST dataset was loaded into the environment using Pandas, containing grayscale images of handwritten digits.

Data Preprocessing

Pixel values were separated from labels and normalized from the range **0–255 to 0–1** to improve model performance and stability.

Noise Addition

Random noise was added to the images to simulate corrupted input data, enabling the model to learn noise removal.

Data Splitting

The dataset was divided into training and testing sets to ensure proper evaluation of the model.

Autoencoder Architecture

An autoencoder model was built consisting of:

- **Encoder** → Compresses input into a lower-dimensional representation using Dense layers
- **Decoder** → Reconstructs the original image from the compressed representation using Dense layers

Model Compilation

The model was compiled using the **Adam optimizer** and **Mean Squared Error (MSE)** loss function, suitable for reconstruction tasks.

Model Training

The autoencoder was trained for multiple epochs, allowing it to learn how to reconstruct clean images from noisy inputs.

Output Generation

After training, the model generated **denoised images** by passing noisy images through the encoder and decoder, effectively removing noise and restoring the original digit patterns.

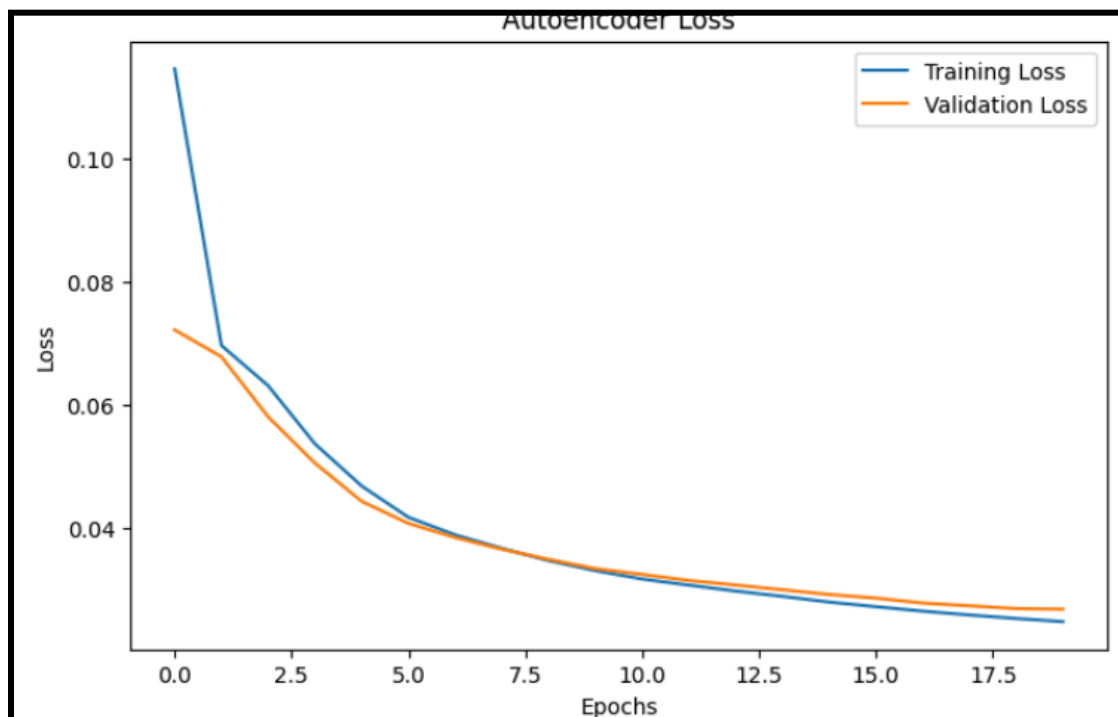
Result

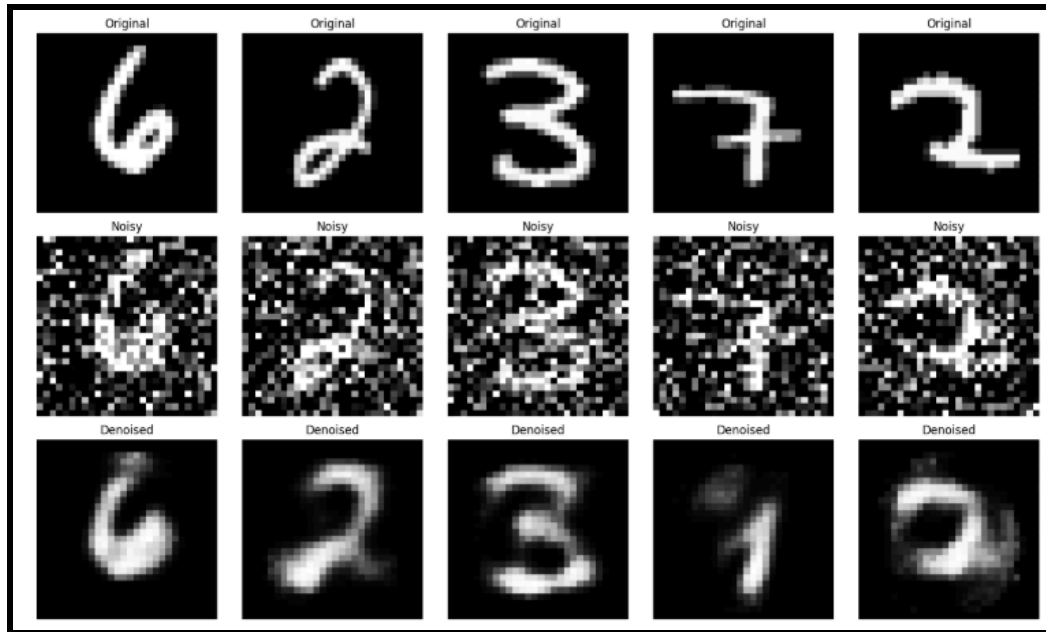
- **Loss (MSE):** Decreased over epochs
- Autoencoder successfully reconstructed clean images

Performance Metrics:

- Mean Squared Error (MSE)

Model Graphs:





Conclusion

The autoencoder effectively removed noise from images by learning to reconstruct clean outputs from corrupted inputs through its encoder–decoder architecture. During training, the model minimized reconstruction error, allowing it to distinguish between noise and important image features. As a result, it successfully generated clearer and more accurate versions of the input images. Additionally, the model learned meaningful and compressed representations of the images in its latent space, capturing essential patterns such as shapes and structures of digits. This ability makes autoencoders highly useful not only for image denoising but also for tasks like image compression, feature extraction, and general image restoration applications.